Making Compassionate Use More Useful: Using real-world data, real-world evidence and digital twins to supplement or supplant randomized controlled trials.

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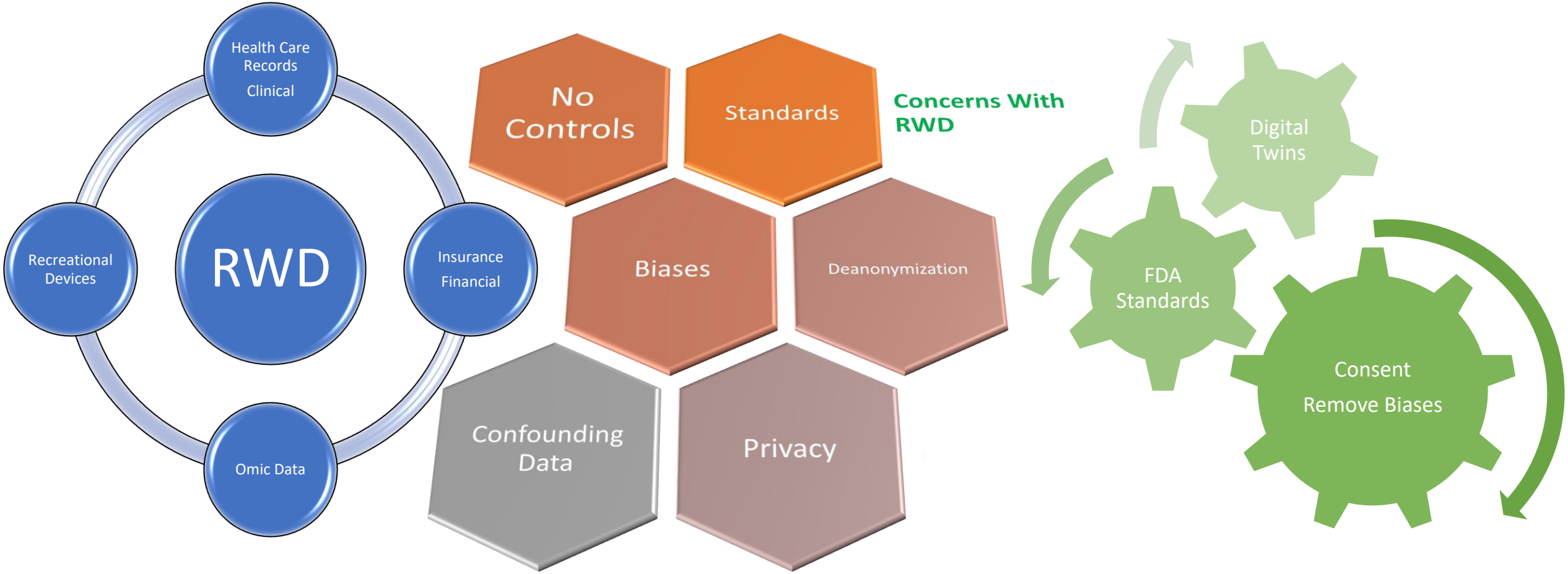
The coronavirus pandemic has placed renewed focus on expanded access (EA) programs to provide compassionate use exceptions to the waves of patients seeking medical care in treating the novel disease. While commendable, justifiable, and compassionate, EA programs are not designed to collect the necessary vital clinical data that can be later used in the New Drug Application process before the U.S. Food and Drug Administration (FDA). In particular, they lack the necessary rigor of properly crafted and controlled randomized controlled trials (RCT) which ensure that each patient closely monitored for side effects and other potential dangers associated with the drug, that the data is documented, stable and are traceable and that the patient population is well defined with the defined target condition. Overall, while RCTs is deemed to be of the most reliable methodologies within evidence-based medicine, morally, however, they are problematic in EA programs. Nevertheless, actionable data ought to be collected from EA patients. To this end, we look to the growing incorporation of real-world data real-world evidence as increasingly useful substitutes for data collected via RCTs, including the ethical, legal and social implications thereof. Finally, we suggest the use of digital twins as an additional method to derive causal inferences from real-world trials involving expanded access patients.